Advanced Causal Inference

Lecture 12: Graphical Causal Models

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This lecture note is partially based on the lecture notes/textbook of Ying Zhou, Robin Evans [Link], Thomas Richardson [Link]

Outline

1. Review
2. Basic structures in a DAG
3. Use of DAGs
4. Relationships between Graphs and Causal Inference Frameworks

Causal Assumptions in Observational Studies

Assumption (SUTVA (Rubin, 1980))

- $Y = Y(A) = AY(1) + (1 - A)Y(0)$

Assumption (Conditional Ignorability)

- $Y(a) \perp\!\!\!\perp A \,|\, X,\ \forall a$

Assumption (Positivity/Overlap)

- $\Pr(A = a \,|\, X) \in (0, 1),\ \forall (a, X)$

- Under these assumptions,

$$\begin{aligned}
\tau_{\mathsf{ATE}} &= \mathrm{E}\{Y(1) - Y(0)\} \\
&= \mathrm{E}\big\{\mathrm{E}(Y|A = 1, X) - \mathrm{E}(Y|A = 0, X)\big\} && \text{OR-based Representation} \\
&= \mathrm{E}\left\{\frac{AY}{\Pr(A = 1|X)} - \frac{(1 - A)Y}{\Pr(A = 0|X)}\right\} && \text{IPW-based Representation}
\end{aligned}$$

Adjustment Set X

- Under these causal assumptions,

$$\text{Association (conditional on } X) = \text{Causation}$$

- However, we have been somewhat implicit about which variables to include in the adjustment set X

- What kind of variables shall we include/exclude in the adjustment set X?

Graphical Causal Models

- **Tool: Graphical causal models**
  - Express knowledge in terms of a directed acyclic graph and a single world intervention graph
- We shall use this tool to answer the following questions:
  - **Question 1:** When does (conditional) association equal causation?
  - **Question 2:** Are two random variables independent after conditioning on a third random variable?
  - **Question 3:** What variables shall we include in a prediction model?

Outline

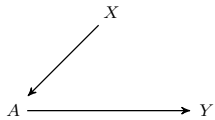
1 Review

2 Basic structures in a DAG

3 Use of DAGs

4 Relationships between Graphs and Causal Inference Frameworks

Directed Acyclic Graphs (DAGs)

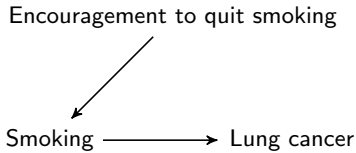


- **Directed arrows** $A \to Y$: A may directly cause Y (but Y may not cause A)
- Absence of arrows between X and Y: X may not directly cause Y (and Y may not directly cause X)

$$\begin{array}{rrcl} & \text{Absence of an arrow} & \Rightarrow & \text{Absence of a causal effect} \\ \Leftrightarrow & \text{Existence of a causal effect} & \Rightarrow & \text{Existence of an arrow} \\ \not\Leftrightarrow & \text{Existence of an arrow} & \Rightarrow & \text{Existence of a causal effect} \end{array}$$

- Implicit assumption: for all pairs of variables on the DAG, their common causes are also on the DAG
- **Acyclic**: no cycles, i.e., $A \to B \to C \to A$ cannot appear
  - This is because cause must precede effect
  - It is possible that $A_t \to Y_t \to A_{t+1}$

Example



- What assumptions are encoded by this DAG?
- Do you think these assumptions are reasonable?

Three Basic Structures in a DAG: Chain



1. Chain

Suppose we are interested in the causal effect of A on C:

- Causation$(A \to C) = A \to B \to C$
- Association$(A, C) = A \to B \to C$
- Association$(A, C \mid B) = 0$ (or equivalently, $A \perp\!\!\!\perp C \mid B$)

Three Basic Structures in a DAG: Chain



1. Chain

Example:

- $A =$ Amount of effort
- $B =$ Course grade
- $C =$ Future success



Conclusion: Do not condition on B (mediator)

Three Basic Structures in a DAG: Chain



1. Chain

Example:

- $A =$ drug
- $B =$ post-treatment blood pressure
- $C =$ recovery



Conclusion: Do not condition on B (mediator)

Three Basic Structures in a DAG: Fork



2. Fork

Suppose we are interested in the causal effect of A on C:

- Causation$(A \rightarrow C) = 0$
- Association$(A, C) = A \leftarrow B \rightarrow C$
- Association$(A, C \mid B) = 0$

Three Basic Structures in a DAG: Fork

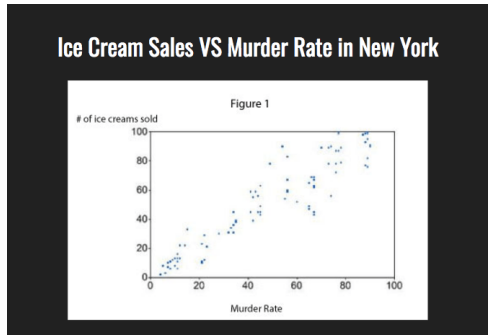
$$A \longleftarrow B \longrightarrow C$$

2. Fork

Example:

- $A =$ ice cream sales
- $B =$ temperature
- $C =$ murder rate

Conclusion: Condition on B (Confounder)



Three Basic Structures in a DAG: Collider



3. Collider

Suppose we are interested in the causal effect of A on C:

- Causation$(A \to C) = 0$
- Association$(A, C) = 0$
- Association$(A, C \mid B) \neq 0$

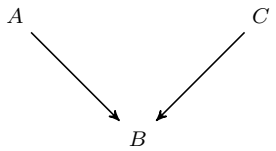
Three Basic Structures in a DAG: Collider



Why would two independent variables suddenly become dependent when we condition on their common effect?

- Condition on B: compare A and C when B takes the same value
- B depends on both A and C
- Any change in A must be compensated for by a change in C

Examples of Collider: Toy Example

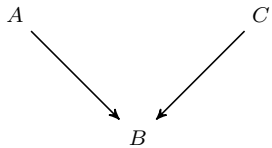


Consider for example:

$$A, C \overset{iid}{\sim} \text{Ber}(0.5) \quad \text{and} \quad B = A \times C$$

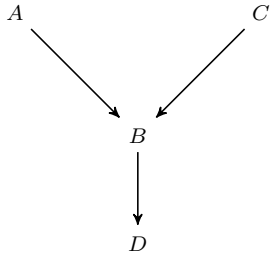
- $A \perp\!\!\!\perp C$ by i.i.d.
- $A \not\perp\!\!\!\perp C \mid B$ because $\Pr(A = 1|B = 0) = \Pr(C = 1|B = 0) = 1/3$ but $\Pr(A = 1, C = 1|B = 0) = 0$

Examples of Collider: Genotype



- Let A and C denote the blood types of the mother and father, respectively
- If we know the blood type of their child, say $B = \text{AB}$,
  we can deduce that neither the mother nor the father has blood type O

A More Complicated Example

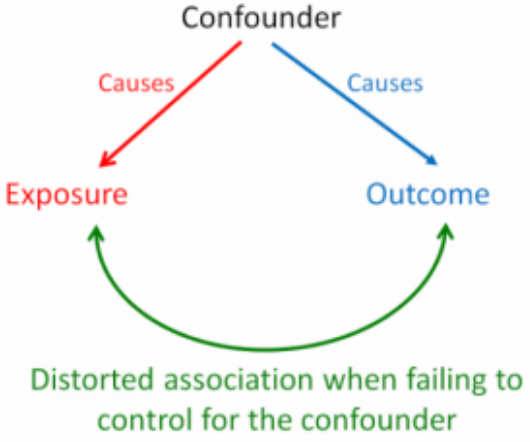


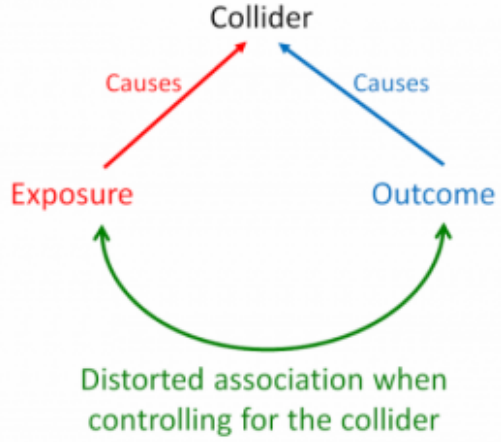
- In real-world examples, the variable B may be unobserved, while only D is observed
- People might then be tempted to use D as a confounder
- But, we have $A \not\perp\!\!\!\perp C \mid D$
- This is a point often overlooked even for people familiar with the notion of collider bias

Implication of a Collider

- **Causally:** Do not conditional on B (collider)
- Collider bias has been surprising to many people
- Reason: humans tend to think association must be a consequence of some sort of causation
- WRONG!

Comparison


Confounder
Causes
Causes
Exposure
Outcome
Distorted association when failing to control for the confounder


Collider
Causes
Causes
Exposure
Outcome
Distorted association when controlling for the collider

What If We Are Interested in Prediction?

- Suppose the true graphical structure is a chain/fork/collider, and we are interested in predicting C using a linear model

- Suppose that we already included B in the model. Can we get a better prediction if we also include A in the model?

  - Chain: $A \perp\!\!\!\perp C \,|\, B$ $\Rightarrow$ including A does not improve the prediction of C
  - Fork: $A \perp\!\!\!\perp C \,|\, B$ $\Rightarrow$ including A does not improve the prediction of C
  - Collider: $A \not\perp\!\!\!\perp C \,|\, B$ $\Rightarrow$ including A **does** improve the prediction of C

Summary

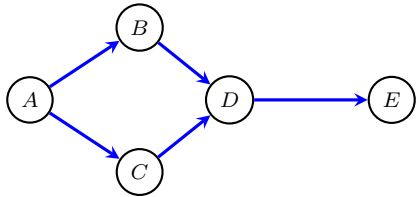
	Causal($A \to C$)	$A \perp\!\!\!\perp C$	$A \perp\!\!\!\perp C \mid B$	Predict C
Chain: $A \to B \to C$	Asso(A, C)	No†	Yes	B
Fork: $A \leftarrow B \to C$	Asso$(A, C \mid B)$	No†	Yes	B
Collider: $A \to B \leftarrow C$	Asso$(A, C) = 0$	Yes	No†	A, B

† : See the slides on faithful DAGs

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Terminologies

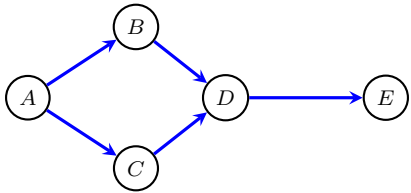


- If there is an edge $A \to B$ then A is said to be a parent of B, and B is a child of A.
- Two nodes joined by an edge are said to be adjacent
- The set of parents of a node A, is denoted by Pa(A)

  Example: Pa$(D) = \{B, C\}$

Ancestors and Descendants

- X is said to be an ancestor of Y if there is a directed path $X \to \cdots \to Y$, or if $X = Y$.
- If X is an ancestor of Y then Y is said to be a descendant of X.
- ancestor vs parent



$$\mathsf{An}(E) = \{A, B, C, D, E\}, \quad \mathsf{An}(D) = \{A, B, C, D\},$$
$$\mathsf{De}(A) = \{A, B, C, D, E\}, \quad \mathsf{De}(B) = \{B, D, E\}.$$

Determine the Adjustment Set for Causal Inference

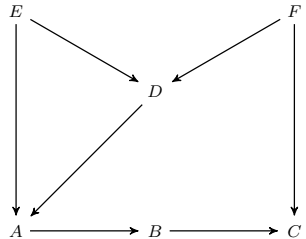
- Goal of adjustment for causal inference:
  - Block all non-causal paths
  - Open all causal paths

- A path from A to C consists of a sequence of distinct nodes that are adjacent

- Causal path from A to C: directed path from A to C, i.e., $A \to \cdots \to C$
  - Example: $A \to C, A \to B \to C, A \to B \to D \to C$

- Non-causal path between A and C: Everything else
  - Example: $A \leftarrow B \to C, \quad A \to B \to D \leftarrow C$

Example



- Any causal path going from A to C?
  - $A \to B \to C$

- Any non-causal path between A and C?
  - $A \leftarrow E \to D \leftarrow F \to C$ and $A \leftarrow D \leftarrow F \to C$

Open vs Blocked Path

A non-endpoint node B on a path (between A and C) is a collider if the path takes the form $A \cdots \rightarrow B \leftarrow \cdots C$

A blocked path is a path that contains a collider

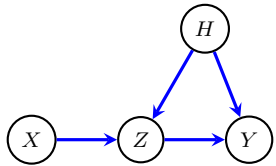
- Example: $A \rightarrow B \rightarrow D \leftarrow C, \quad A \rightarrow B \leftarrow D \rightarrow E \leftarrow C$

An open path is a path that contains no colliders

- Example: $A \rightarrow B \rightarrow C, \quad A \leftarrow B \rightarrow D \rightarrow C$
- All causal paths are open (unconditionally)

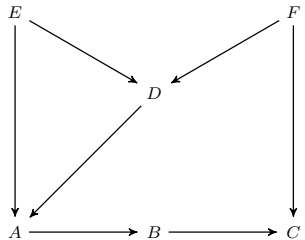
Subtlety Regarding Colliders

- A node is a collider or non-collider *with respect to* a given path
- A node may be a collider on one path and a non-collider on another (even between the same endpoints)



- Here Z is a collider on the path $X \to Z \leftarrow H \to Y$,
  but Z is a non-collider on the path $X \to Z \to Y$
- $\ldots$ V is a collider **with respect to the path** $\ldots$

Example (cont.)



- Which of the paths between A and C are open? blocked?
- Causal paths are open:
  - $A \to B \to C$: Open
- Non-causal paths:
  - $A \leftarrow E \to D \leftarrow F \to C$: Blocked due to D
  - $A \leftarrow D \leftarrow F \to C$: Open

Block/Open A Path by Conditioning

We can block an *open* path by conditioning on any non-collider on the path

- Example: Block $A \to B \to C$ by conditioning on B; denoted by $A \to \boxed{B} \to C$

  Block $A \leftarrow B \to D \to C$ by conditioning on B or D or $\{B, D\}$

We can open a *blocked* path by conditioning on (descendants of) all colliders on the path

- Example: Open $A \to B \to D \leftarrow C$ by conditioning on D

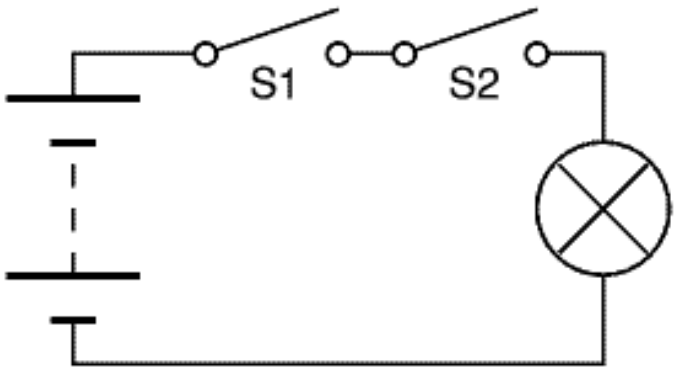
  Open $A \to B \leftarrow D \to E \leftarrow C$ by conditioning on B and E

Question: Is $A \to B \leftarrow D \to E \leftarrow C$ open or blocked if we condition on B? D? E? $\{B, D, E\}$?

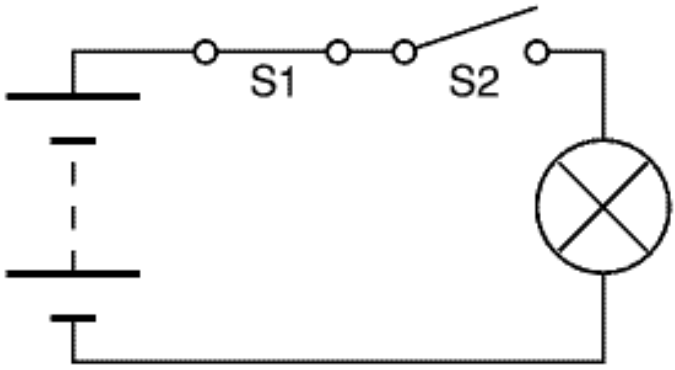
- Condition on B: $A \to \boxed{B} \leftarrow D \to E \leftarrow C$; blocked
- Condition on D: $A \to B \leftarrow \boxed{D} \to E \leftarrow C$; blocked
- Condition on E: $A \to B \leftarrow D \to \boxed{E} \leftarrow C$; blocked
- Condition on B, C, D: $A \to \boxed{B} \leftarrow \boxed{D} \to \boxed{E} \leftarrow C$: blocked

  $A \to \boxed{B} \leftarrow D \to \boxed{E} \leftarrow C$ is open, which can be blocked when any non-collider on the path (i.e. D) is conditioned

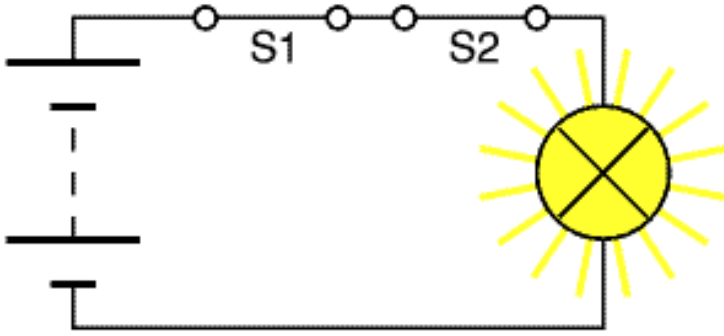
Recall: Series Circuit from Physics



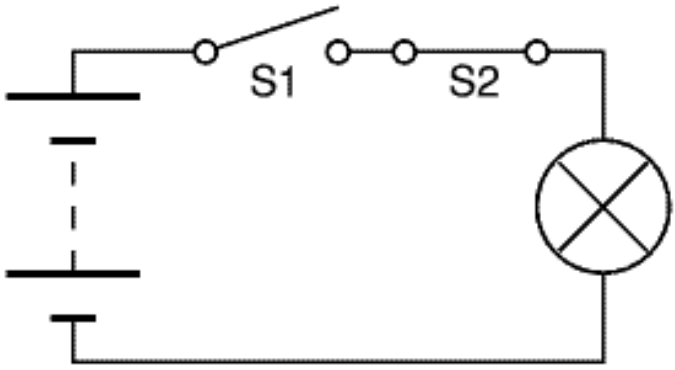
Recall: Series Circuit from Physics



Recall: Series Circuit from Physics



Recall: Series Circuit from Physics



Path = Series Circuit

Rule 1: A series circuit is complete if all the switches are ON

Example: $A \rightarrow B \leftarrow D \rightarrow E \leftarrow C$

- Collider = Switch is OFF
- Non-collider = Switch is ON

Rule 2: Conditioning on a node = change the status of the switch

- Condition on a collider = Turn on the switch
- Condition on a non-collider = Turn off the switch

Note: condition on any descendant of a collider turns on the switch

Recall

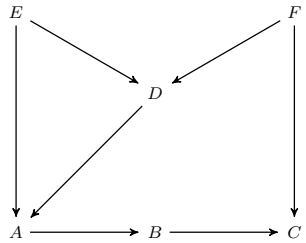
Recall: **Goal** of adjustment for causal inference:

- Block all non-causal paths
- Open all causal paths

so that

$$\text{Conditional association} = \text{Causation}$$

Example (cont.)



- Find a set of variables so that conditional on which, association between A and C equals causation

- Causal paths are open:
  - $A \to B \to C$: Open $\Rightarrow$ don't change any variables on this path

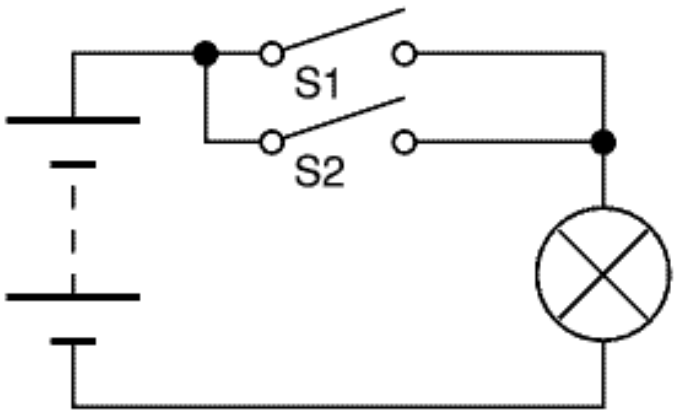
- Non-causal paths:
  - $A \leftarrow E \to D \leftarrow F \to C$: Blocked $\Rightarrow$ don't change any variables on this path (esp. do not condition on D)
  - $A \leftarrow D \leftarrow F \to C$: Open $\Rightarrow$ need to block this path $\Rightarrow$ condition on F (why not D?)

D-separation

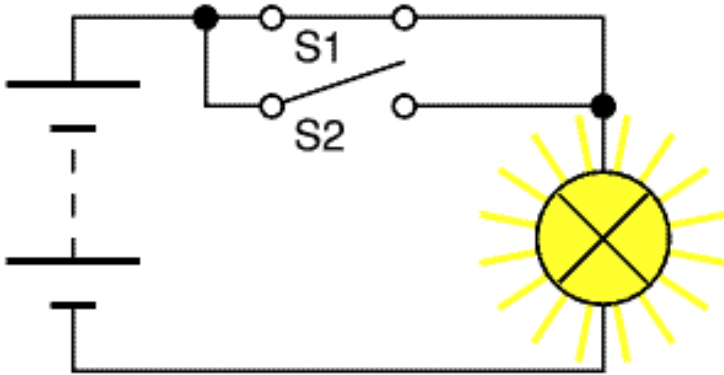
Rule 3: D-separation (D=dependence)

- Two nodes are d-connected (likely dependent) if at least one of the paths between them is open
- Two nodes are d-separated (independent) if all the paths between them are blocked

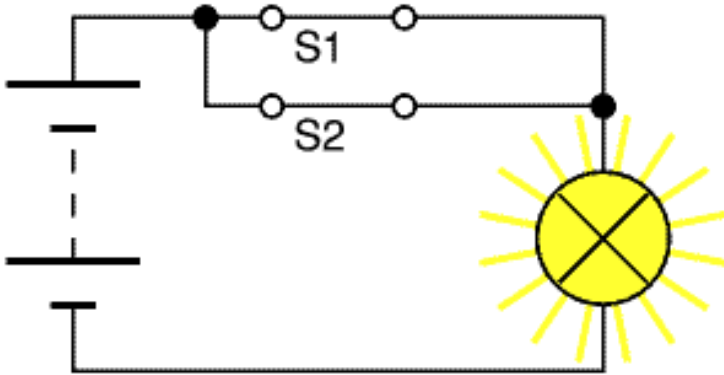
Recall: Parallel Circuit from Physics



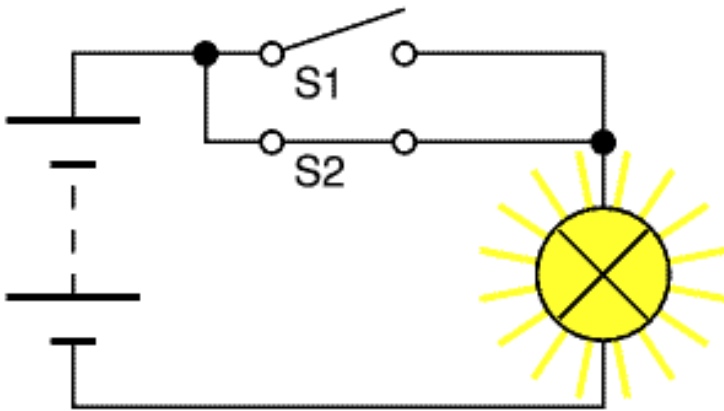
Recall: Parallel Circuit from Physics



Recall: Parallel Circuit from Physics



Recall: Parallel Circuit from Physics



DAG $\approx$ Parallel Circuit

Rule 3: A parallel circuit is complete if and only if at least one of the paths is complete

- Two nodes are d-connected (likely dependent) if at least one of the paths between them is open
- Two nodes are d-separated (independent) if all the paths between them are blocked

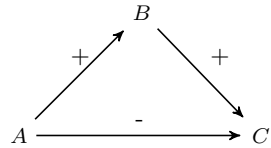
Unlikely Cases

- When are two d-connected nodes independent?

1. $A \perp\!\!\!\perp C$ if the association through $A \to C$ happens to cancel out with the association through $A \to B \to C$

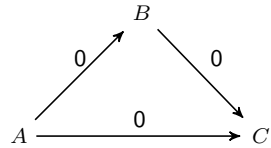
$$A = \epsilon_A$$
$$B = 0.5A + \epsilon_B$$
$$C = 2B - A + \epsilon_C = 2\epsilon_B + \epsilon_C \perp\!\!\!\perp \epsilon_A$$

   - Won't happen in parallel circuits as the current cannot be negative!



2. $A \perp\!\!\!\perp C$ if there is no association but arrows are present

   - Recall that an arrow from A to B means A may directly cause B



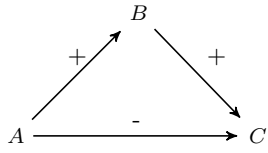
Unlikely Cases

- When are two d-connected nodes independent?
- $A \perp\!\!\!\perp C$ if the association through $A \to C$ happens to cancel out with the association through $A \to B \to C$

$$A = \epsilon_A$$

$$B = 0.5A + \epsilon_B$$

$$C = 2B - A + \epsilon_C = 2\epsilon_B + \epsilon_C \perp\!\!\!\perp \epsilon_A$$



  - Won't happen in parallel circuits as the current cannot be negative!

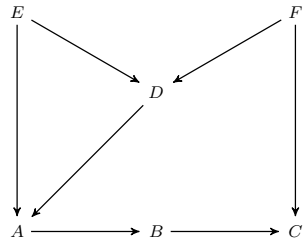
Faithful DAGs

We say a DAG is faithful if

$$A \text{ is d-connected with } B \text{ conditional on } C \quad \Leftrightarrow \quad A \not\perp\!\!\!\perp B \mid C$$

Example (cont.)

- Suppose the DAG is faithful. Which of the following is true?
  - $A \perp\!\!\!\perp C \mid F$
  - $A \perp\!\!\!\perp C \mid B, D$
  - $E \perp\!\!\!\perp F \mid A$



1. Open path $A \rightarrow B \rightarrow C$
2. Open path $A \leftarrow E \rightarrow \boxed{D} \leftarrow F \rightarrow C$
3. Open path $E \rightarrow \boxed{A} \leftarrow D \leftarrow F$

Algorithm

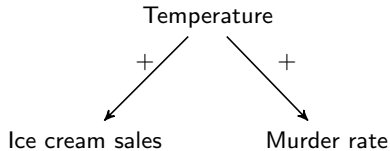
Algorithm 1: An algorithm to decide if $A \perp\!\!\!\perp C \mid B$ from a DAG

List all paths connecting A and B, say $p_1, \ldots, p_K$;
for $i = 1, \ldots, K$ **do**
  if p_i *is open & the DAG is faithful* **then**
    Return: $A \not\perp\!\!\!\perp B \mid C$
  end
  ;
  if p_i *is open & the DAG is not faithful* **then**
    Return: Insufficient information;
  end
  if p_i *is blocked* **then**
    Go to the next path, i.e., $i \leftarrow i + 1$
  end
end
if *all paths are blocked* **then**
  Return: $A \perp\!\!\!\perp B \mid C$
end

Rule for Prediction

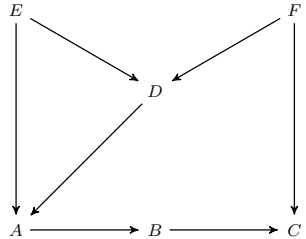
If $A \perp\!\!\!\perp C \mid B$, then there is no need to include A in the prediction model for C (as long as B is already in the model)

Example:



- Under faithfulness, $A \perp\!\!\!\perp C \mid B$ implies there is no open path between A and C conditional on B
- In particular, this implies A has no direct path to C. Why?
- If there were $A \leftarrow C$ (or $A \rightarrow C$), then A and C would be d-connected conditional on B, which would also imply $A \not\perp\!\!\!\perp C|B$ in a faithful DAG—a contradiction

Markov Blanket



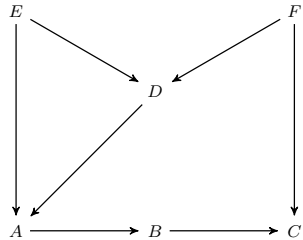
Markov blanket = Parents ∪ Children ∪ Spouses;

- Spouse = nodes sharing a common child but have no direct edge connecting them

Theorem: A node is independent of all other nodes, conditional on its Markov blanket, i.e.,

$$P\Big\{C \,\Big|\, \text{all other nodes}\Big\} = P\Big\{C \,\Big|\, \text{Markov Blanket}(C)\Big\}$$

Example (cont.)



- Suppose I want to predict B. What variables shall I include in the prediction model?
- Note that: Parent $= A$; Children $= C$; Spouse $= F$
- So it suffices to use (A, C, F)

Outline

1 Review

2 Basic structures in a DAG

3 Use of DAGs

4 Relationships between Graphs and Causal Inference Frameworks

Non-Parametric Structural Equation Models

- We start with Pearl (1995)'s NPSEM

- In an NPSEM model associated with a graph $\mathcal{G}$, each variable is given by an equation expressing the variable as a ft of its parents + error term

$$A = f_X(\epsilon_X) \ , \quad M = f_M(A, \epsilon_M) \ , \quad Y = f_Y(M, A, \epsilon_Y)$$

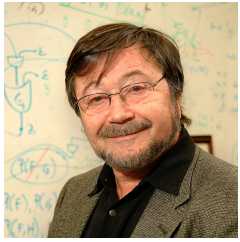
where $\epsilon_X \perp\!\!\!\perp \epsilon_M \perp\!\!\!\perp \epsilon_Y$

- The corresponding potential outcomes are

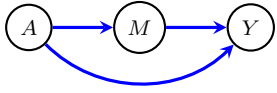
$$A = f_X(\epsilon_X) \ , \quad M(a) = f_M(a, \epsilon_M) \ , \quad Y(a, m) = f_Y(m, a, \epsilon_Y)$$

with the observed variables are defined via SUTVA

$$M = M(A); \quad Y = Y(A, M)$$



Judea Pearl (Citation $>$ 160K)



DAG $\mathcal{G}$

NPSEM-IEs

- Pearl: DAGs and Potential Outcomes are "equivalent theories" (Pearl, 2009; Bollen and Pearl, 2013)
- Pearl's approach to unifying graphs and counterfactuals associates with a DAG the counterfactual model corresponding to an NPSEM **with independent errors** (NPSEM-**IEs**) aka structural causal models
- Surprisingly, the IE component sparks controversy in causal inference, and it drives new causal perspectives

NPSEM-IEs

- Consider a causal model for a binary A and binary Y
- The full data consist of three variables: $O_F = (Y(1), Y(0), A)$
- Therefore, the degrees of freedom for the full data are $2^3 - 1 = 7$
- In order to identify the ATE, we assume ignorability:

$$Y(a) \perp\!\!\!\perp A \text{ for } a = 0, 1 \tag{1}$$

- This assumption imposes two additional constraints, reducing the degrees of freedom of the full data relevant for ATE identification to $7 - 2 = 5$

NPSEM-IEs

- If the nonparametric structural equation model with independent errors (NPSEM-IE) corresponds exactly to the potential outcomes framework, it does not introduce any additional assumptions
- However, we find one more restriction:

$$A = f_A(\epsilon_A)\ , \quad Y(a) = f_Y(a, \epsilon_Y) \quad \Rightarrow \quad \{f_Y(1, \epsilon_Y), f_Y(0, \epsilon_Y)\} \perp\!\!\!\perp \epsilon_A \quad \Rightarrow \quad \{Y(1), Y(0)\} \perp\!\!\!\perp A \tag{2}$$

- Note that $V_1 \perp\!\!\!\perp V_3$ and $V_2 \perp\!\!\!\perp V_3$ does not imply $(V_1, V_2) \perp\!\!\!\perp V_3$
- Therefore, (3) introduces an additional restriction beyond (1), reducing the degrees of freedom of the full data to $5 - 1 = 4$
- In other words, the NPSEM-IE is overly restrictive, imposing stronger assumptions than are necessary

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- In other words, the NPSEM-IE is overly restrictive, imposing stronger assumptions than are necessary

- In fact, these restrictions arise from the assumption of independent errors

- This assumption cannot be verified from any randomized experiment

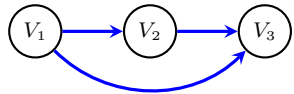
Finest Fully Randomized Causally Interpretable Structured Tree Graph (FFRCISTG)

- Robins (1986); Robins and Richardson (2011)

- Given a DAG $\mathcal{G}$ with node set V, a FFRCISTG model associated with $\mathcal{G}$ makes four assumptions

1. All one-step ahead counterfactuals $V_m(\overline{v}_{m-1})$ exist for any intervention level $\overline{v}_{m-1} \in \overline{V}_{m-1}$ of its predecessors
2. $V_m(\overline{v}_{m-1})$ only depends on V_m's parents, i.e., $V_m(\overline{v}_{m-1}) = V_m(\text{Pa}_m)$
3. Both the factual/counterfactual variables $(V_m, V_m(r))$ for any $R \subseteq V$ are obtained recursively from $V_m(\overline{v}_{m-1})$
4. The following independence holds:

$$\{V_{m+1}(\overline{v}_m), \ldots, V_M(\overline{v}_{M-1})\} \perp\!\!\!\perp V_m(\overline{v}_{m-1}) \,\big|\, \overline{V}_{m-1} = \overline{v}_{m-1}$$

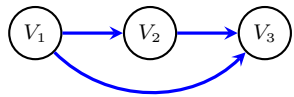


DAG $\mathcal{G}$

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DAG $\mathcal{G}$

- This implies

Counterfactual	Factual
$V_1(\varnothing)$	V_1
$V_2(v_1)$	$V_2 = V_2(V_1)$
$V_3(v_1, v_2)$	$V_3 = V_3\big(V_1, V_2(V_1)\big)$

and

$$\big\{V_3(v_1 = 1, v_2), V_2(v_1 = 1)\big\} \perp\!\!\!\perp V_1 \quad , \quad \big\{V_3(v_1 = 0, v_2), V_2(v_1 = 0)\big\} \perp\!\!\!\perp V_1$$
$$V_3(v_1 = 1, v_2) \perp\!\!\!\perp V_2(v_1 = 1) \,\big|\, V_1 = 1 \quad , \quad V_3(v_1 = 0, v_2) \perp\!\!\!\perp V_2(v_1 = 0) \,\big|\, V_1 = 0$$

FFRCISTG vs. NPSEM-IE

- Note that a FFRCISTG model does not introduce the so-called **cross-world** assumptions

- In the previous example's FFRCISTG model does not impose

$$\{V_2(v_1 = 0), V_2(v_1 = 1), V_3(v_1 = 0, v_2 = 0), \ldots, V_3(v_1 = 1, v_2 = 1)\} \perp\!\!\!\perp V_1$$

$$V_3(v_1 = 1, v_2) \perp\!\!\!\perp V_2(v_1 = 0) \,\big|\, V_1 = 1 \quad , \quad V_3(v_1 = 1, v_2) \perp\!\!\!\perp V_2(v_1 = 0) \,\big|\, V_1 = 0$$

- The NPSEM-IEs actually implies this restriction, unlike FFRCISTGs

- Practically, FFRCISTGs + Cross-world assumptions = NPSEM-IEs

- But, FFRCISTG models are difficult to understand compared to NPSEM-IEs

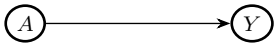
Single World Intervention Graph (SWIG)

- Richardson and Robins (2013) introduced SWIGs to unite the potential outcome and graphical approaches to causal inference

- SWIGs achieve this by explicitly including the potential outcomes in the graph based on a FFRCISTG model

DAG vs. SWIG

- Suppose $Y(1) \perp\!\!\!\perp A$ and $Y(0) \perp\!\!\!\perp A$ (note, this does not imply $\{Y(1), Y(0)\} \perp\!\!\!\perp A$)
- A DAG corresponding to this assumption

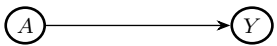


- Somewhat abstract, as it does not include any potential outcomes
- Unclear whether the assumption $Y(a) \perp\!\!\!\perp A$ is encoded

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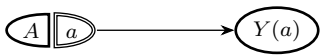


- Somewhat abstract, as it does not include any potential outcomes

- Unclear whether the assumption $Y(a) \perp\!\!\!\perp A$ is encoded

- A single DAG may correspond to multiple SWIGs i.e., a DAG is a coarsened version of a SWIG

- A SWIG corresponding to this assumption



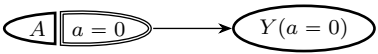
- Clearly represent potential outcomes in the graph

- The assumption $Y(a) \perp\!\!\!\perp A$ is encoded in the graph

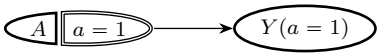
SWIG = Graph-valued Function

- SWIG $\mathcal{G}(\cdot)$ is a graph-valued function:

$$\mathcal{G}(\cdot) : \{0, 1\} \to \text{SWIG} \quad \overset{1-1}{\leftrightarrow} \quad \text{identifiable law}$$



SWIG $\mathcal{G}(a = 0)$



SWIG $\mathcal{G}(a = 1)$

- Takes a treatment level a as an input
- Returns a SWIG $\mathcal{G}(a)$ as an output

Construction of a SWIG

- Given a graph $\mathcal{G}$, a subset of vertices $\mathcal{A} = \{A_1, \ldots, A_k\}$ to be intervened on, we form $\mathcal{G}(a)$ in two steps:

1. (Node splitting): For every $A \in \mathcal{A}$, split the node into a random node A and a fixed node a



- The random half inherits all edges directed into A; the fixed half inherits all edges directed out of A

2. Relabel descendants of fixed nodes



Property of SWIGs

- Original graph $\mathcal{G}$: observed distribution $P(\boldsymbol{V})$ versus SWIG $\mathcal{G}(\boldsymbol{a})$: counterfactual distribution $P(\boldsymbol{V}(\boldsymbol{a}))$

- **Factorization of counterfactual variables**:

  Distribution $P(\boldsymbol{V}(\boldsymbol{a}))$ over the variables in $\mathcal{G}(\boldsymbol{a})$ factorizes with respect to the SWIG $\mathcal{G}(\boldsymbol{a})$ (ignoring fixed nodes)

- **Modularity**: $P(\boldsymbol{V}(\boldsymbol{a}))$ and $P(\boldsymbol{V})$ are linked as follows:

  conditional density associated with $Y(\boldsymbol{a})$ in $\mathcal{G}(\boldsymbol{a})$

  $=$ conditional density associated with Y in $\mathcal{G}$ after substituting $\boldsymbol{a}$ for any $A_i \in \boldsymbol{A} \cap \mathrm{Pa}(Y)$

- **Consequence**: If $P(\boldsymbol{V})$ is observed, then $P(\boldsymbol{V}(\boldsymbol{a}))$ is identified

- **d-separation**:
  - A fixed node is always blocked if it occurs as a non-endpoint on a path
  - A path on which one endpoint is a fixed node can d-connect that node to a random node if it satisfies the usual conditions on colliders and non-colliders

Examples

- Example 1: Randomized experiments



SWIG $\mathcal{G}(a=0)$



SWIG $\mathcal{G}(a=1)$

- $\mathcal{G}(0)$ represents $P\{Y(0), A\} = P\{Y(0)\}P(A)$
- $\mathcal{G}(1)$ represents $P\{Y(1), A\} = P\{Y(1)\}P(A)$

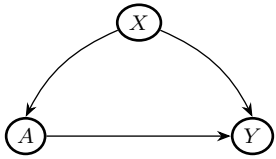
- Under no confounding these marginals are identified from $P(A, Y)$

- However, $Y(0), Y(1)$ are never on the same SWIG, reflecting that $P\{Y(1), Y(0), A\}$ is not identified

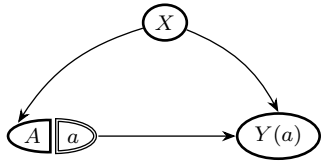
- **Single-world** intervention graph

Examples

- Example 2: Confounding



$\mathcal{G}$



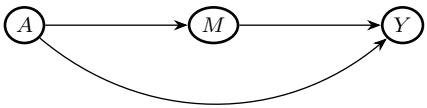
$\mathcal{G}(a)$

- Here we can read directly from the template that

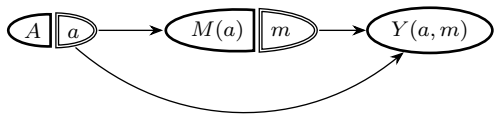
$$Y(a) \perp\!\!\!\perp A|X$$

Examples

- Example 3: Mediation



$\mathcal{G}$



$\mathcal{G}(a)$

- Here we can read directly from the template that

$$\{M(a), Y(a,m)\} \perp\!\!\!\perp A$$

$$M(a) \perp\!\!\!\perp Y(a,m) \,|\, A = a$$

Spoiler: Role of SWIGs in Mediation

- We will revisit SWIGs to clarify the contrasting perspectives of Pearl's mediation framework and Robins and Richardson's treatment decomposition framework

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